Guía Completa de Ejercicios: Métodos de Integración

Cálculo de una Variable

Tarea 2

Cálculo Integral: Métodos de Integración

Modalidad: Equipos de máximo 6 integrantes.
Día de entrega: Pendiente.

1. Cambio de Variable (Sustitución)

Ejercicios

Nivel Medio-Bajo

1. $\displaystyle\int x\sqrt{x^2+5}\,dx$
2. $\displaystyle\int \frac{\ln(x)}{x}\,dx$
3. $\displaystyle\int e^{3x-1}\,dx$

Nivel Medio

4. $\displaystyle\int \frac{x^2}{(x^3+1)^2}\,dx$
5. $\displaystyle\int \frac{\cos(\sqrt{x})}{\sqrt{x}}\,dx$
6. $\displaystyle\int \frac{e^x}{1+e^{2x}}\,dx$
7. $\displaystyle\int \frac{x}{\sqrt{1-x^4}}\,dx$

Nivel Medio-Alto

8. $\displaystyle\int x^3\sqrt{x^2+1}\,dx$
9. $\displaystyle\int \frac{1}{x\sqrt{\ln x-1}}\,dx$
10. $\displaystyle\int \frac{\sin x\cos x}{\sqrt{1+\sin^2 x}}\,dx$

Soluciones

1. $u = x^2 + 5 \implies \frac{1}{3}(x^2+5)^{3/2} + C$
2. $u = \ln x \implies \frac{1}{2}(\ln x)^2 + C$
3. $u = 3x - 1 \implies \frac{1}{3}e^{3x-1} + C$
4. $u = x^3 + 1 \implies -\frac{1}{3(x^3+1)} + C$
5. $u = \sqrt{x} \implies 2\sin(\sqrt{x}) + C$
6. $u = e^x \implies \arctan(e^x) + C$
7. $u = x^2 \implies \frac{1}{2}\arcsin(x^2) + C$
8. $u = x^2 + 1 \implies \frac{1}{5}(x^2+1)^{5/2} - \frac{1}{3}(x^2+1)^{3/2} + C$
9. $u = \ln x - 1 \implies 2\sqrt{\ln x - 1} + C$
10. $u = 1 + \sin^2 x \implies \sqrt{1 + \sin^2 x} + C$

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2. Integración por Partes

Ejercicios

Nivel Medio-Bajo

1. $\int xe^x \, dx$
2. $\int x \cos x \, dx$
3. $\int \ln x \, dx$

Nivel Medio

4. $\int x^2 \sin x \, dx$
5. $\int \arcsin x \, dx$
6. $\int x^3 e^{x^2} \, dx$
7. $\int x \sec^2 x \, dx$

Nivel Medio-Alto

8. $\int e^x \cos x \, dx$
9. $\int (\ln x)^2 \, dx$
10. $\int x^2 \arctan x \, dx$

Soluciones

1. $u = x$, $dv = e^x dx \implies xe^x - e^x + C$
2. $u = x$, $dv = \cos x dx \implies x \sin x + \cos x + C$
3. $u = \ln x$, $dv = dx \implies x \ln x - x + C$
4. $u = x^2$ (dos veces por partes) $\implies -x^2 \cos x + 2x \sin x + 2 \cos x + C$
5. $u = \arcsin x$, $dv = dx \implies x \arcsin x + \sqrt{1 - x^2} + C$
6. $u = x^2$, $dv = xe^{x^2} dx \implies \frac{1}{2} x^2 e^{x^2} - \frac{1}{2} e^{x^2} + C$
7. $u = x$, $dv = \sec^2 x dx \implies x \tan x + \ln |\cos x| + C$
8. Integral cíclica $\implies \frac{e^x(\sin x + \cos x)}{2} + C$
9. Dos veces por partes $\implies x(\ln x)^2 - 2x \ln x + 2x + C$
10. $u = \arctan x$, $dv = x^2 dx \implies \frac{x^3}{3} \arctan x - \frac{x^2}{6} + \frac{1}{6} \ln(1 + x^2) + C$

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3. Fracciones Parciales

Ejercicios

Nivel Medio-Bajo

1. $\displaystyle\int \frac{1}{x^2 - 4} \, dx$
2. $\displaystyle\int \frac{x + 5}{x^2 + 3x + 2} \, dx$
3. $\displaystyle\int \frac{2x - 1}{x^2 - x - 6} \, dx$

Nivel Medio

4. $\displaystyle\int \frac{x^2 + 1}{x^3 - x} \, dx$
5. $\displaystyle\int \frac{3x^2 - 2x + 1}{(x - 1)^2 (x + 1)} \, dx$
6. $\displaystyle\int \frac{x^3}{x^2 + 2x + 1} \, dx$
7. $\displaystyle\int \frac{2x^2 + 3}{(x^2 + 1)(x - 1)} \, dx$

Nivel Medio-Alto

8. $\displaystyle\int \frac{1}{x^3 + 1} \, dx$
9. $\displaystyle\int \frac{x^4}{x^4 - 1} \, dx$
10. $\displaystyle\int \frac{x^2 - 1}{(x^2 + 1)(x^2 + 4)} \, dx$

Soluciones

1. $\frac{1}{4}\ln\left|\frac{x-2}{x+2}\right| + C$
2. $4\ln|x+1| - 3\ln|x+2| + C$
3. $\ln|(x-3)(x+2)| + C$
4. $\ln\left|\frac{x^2-1}{x}\right| + C$
5. $\ln|x-1| - \frac{1}{x-1} + 2\ln|x+1| + C$
6. División sintética previa $\implies \frac{x^2}{2} - 2x + 3\ln|x+1| + \frac{1}{x+1} + C$
7. $\frac{5}{2}\ln|x-1| - \frac{1}{4}\ln(x^2+1) - \frac{1}{2}\arctan x + C$
8. Factorización $x^3+1 = (x+1)(x^2-x+1) \implies \frac{1}{3}\ln|x+1| - \frac{1}{6}\ln(x^2-x+1) + \frac{1}{\sqrt{3}}\arctan\left(\frac{2x-1}{\sqrt{3}}\right) + C$
9. División de polinomios previa $\implies x + \frac{1}{4}\ln\left|\frac{x-1}{x+1}\right| - \frac{1}{2}\arctan x + C$
10. $-\frac{2}{3}\arctan x + \frac{5}{6}\arctan\left(\frac{x}{2}\right) + C$

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4. Integración Trigonométrica

Ejercicios

Nivel Medio-Bajo

1. $\displaystyle\int \sin^3 x\, dx$
2. $\displaystyle\int \cos^2 x\, dx$
3. $\displaystyle\int \tan^2 x\, dx$

Nivel Medio

4. $\displaystyle\int \sin^4 x\, dx$
5. $\displaystyle\int \sin^3 x \cos^2 x\, dx$
6. $\displaystyle\int \tan^3 x \sec x\, dx$
7. $\displaystyle\int \cos(3x)\cos(2x)\, dx$

Nivel Medio-Alto

8. $\displaystyle\int \sec^3 x\, dx$
9. $\displaystyle\int \sin^2 x \cos^4 x\, dx$
10. $\displaystyle\int \frac{\sin^3 x}{\cos^4 x}\, dx$

Soluciones

1. Identidad $\sin^2 x = 1 - \cos^2 x \implies -\cos x + \frac{1}{3}\cos^3 x + C$
2. Ángulo doble $\implies \frac{x}{2} + \frac{\sin 2x}{4} + C$
3. Identidad $\tan^2 x = \sec^2 x - 1 \implies \tan x - x + C$
4. Reducción de potencia $\implies \frac{3}{8}x - \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + C$
5. Identidad $\sin^2 x = 1 - \cos^2 x \implies -\frac{1}{3}\cos^3 x + \frac{1}{5}\cos^5 x + C$
6. $u = \sec x \implies \frac{1}{3}\sec^3 x - \sec x + C$
7. Producto a suma $\implies \frac{1}{2}\sin x + \frac{1}{10}\sin 5x + C$
8. Por partes ($u = \sec x$, $dv = \sec^2 x dx$) $\implies \frac{1}{2}\sec x \tan x + \frac{1}{2}\ln|\sec x + \tan x| + C$
9. Identidades de ángulo doble $\implies \frac{x}{16} + \frac{\sin 2x}{32} - \frac{\sin 4x}{64} - \frac{\sin 6x}{192} + C$
10. $u = \cos x \implies \frac{1}{3}\sec^3 x - \sec x + C$

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5. Sustitución Trigonométrica

Ejercicios

Nivel Medio-Bajo

1. $\displaystyle\int \frac{1}{\sqrt{4 - x^2}}\, dx$
2. $\displaystyle\int \frac{1}{x^2 + 9}\, dx$
3. $\displaystyle\int \sqrt{1 - x^2}\, dx$

Nivel Medio

4. $\displaystyle\int \frac{x^2}{\sqrt{9 - x^2}}\, dx$
5. $\displaystyle\int \frac{1}{x\sqrt{x^2 - 4}}\, dx$
6. $\displaystyle\int \frac{1}{(x^2 + 4)^{3/2}}\, dx$
7. $\displaystyle\int \frac{x^3}{\sqrt{x^2 + 1}}\, dx$

Nivel Medio-Alto

8. $\displaystyle\int \sqrt{x^2 - 9}\, dx$
9. $\displaystyle\int \frac{1}{\sqrt{x^2 + 2x + 5}}\, dx$
10. $\displaystyle\int \frac{\sqrt{x^2 - 4}}{x^3}\, dx$

Soluciones

1. $x = 2\sin\theta \implies \arcsin\left(\frac{x}{2}\right) + C$
2. $x = 3\tan\theta \implies \frac{1}{3}\arctan\left(\frac{x}{3}\right) + C$
3. $x = \sin\theta \implies \frac{1}{2}x\sqrt{1-x^2} + \frac{1}{2}\arcsin x + C$
4. $x = 3\sin\theta \implies \frac{9}{2}\arcsin\left(\frac{x}{3}\right) - \frac{1}{2}x\sqrt{9-x^2} + C$
5. $x = 2\sec\theta \implies \frac{1}{2}\operatorname{arcsec}\left(\frac{x}{2}\right) + C$
6. $x = 2\tan\theta \implies \frac{x}{4\sqrt{x^2+4}} + C$
7. $x = \tan\theta \implies \frac{1}{3}(x^2+1)^{3/2} - \sqrt{x^2+1} + C$
8. $x = 3\sec\theta \implies \frac{x}{2}\sqrt{x^2-9} - \frac{9}{2}\ln\left|x + \sqrt{x^2-9}\right| + C$
9. Completar cuadrado $(x+1)^2 + 4 \implies \ln\left|(x+1) + \sqrt{x^2+2x+5}\right| + C$
10. $x = 2\sec\theta \implies \frac{1}{4}\operatorname{arcsec}\left(\frac{x}{2}\right) - \frac{\sqrt{x^2-4}}{2x^2} + C$

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6. Miscelánea (Combinación de Varios Métodos)

Ejercicios

1. $\displaystyle\int \sqrt{e^x - 1}\,dx$
2. $\displaystyle\int \frac{\arcsin x}{x^2}\,dx$
3. $\displaystyle\int \frac{\ln(x+1)}{\sqrt{x}}\,dx$
4. $\displaystyle\int \frac{1}{1+\sin x+\cos x}\,dx$
5. $\displaystyle\int e^{\sqrt{x}}\,dx$
6. $\displaystyle\int \frac{x\arctan x}{(1+x^2)^2}\,dx$
7. $\displaystyle\int \frac{\cos x}{1+\sin^2 x}\,dx$
8. $\displaystyle\int \frac{1}{x^2\sqrt{x^2+4}}\,dx$
9. $\displaystyle\int \frac{x\ln x}{(x^2-1)^{3/2}}\,dx$
10. $\displaystyle\int \frac{\sec^2 x}{\tan^3 x - \tan x}\,dx$

Soluciones

1. **Sustitución** ($u = \sqrt{e^x - 1}$) + **Fracciones parciales**:
   $2\sqrt{e^x - 1} - 2\arctan(\sqrt{e^x - 1}) + C$

2. **Por partes** ($u = \arcsin x$, $dv = x^{-2}dx$) + **Sustitución trigonométrica**:
   $-\frac{\arcsin x}{x} - \ln\left|\frac{1+\sqrt{1-x^2}}{x}\right| + C$

3. **Por partes** + **Sustitución algebraica**:
   $2\sqrt{x}\ln(x+1) - 4\sqrt{x} + 4\arctan(\sqrt{x}) + C$

4. **Sustitución de Weierstrass** ($t = \tan(x/2)$):
   $\ln\left|1 + \tan\left(\frac{x}{2}\right)\right| + C$

5. **Sustitución** ($t = \sqrt{x}$) + **Por partes**:
   $2\sqrt{x}e^{\sqrt{x}} - 2e^{\sqrt{x}} + C$

6. **Por partes** + **Sustitución trigonométrica**:
   $\frac{(x^2-1)\arctan x + x}{4(1+x^2)} + C$

7. **Sustitución directa** ($u = \sin x$):
   $\arctan(\sin x) + C$

8. **Sustitución trigonométrica** ($x = 2\tan\theta$):
   $-\frac{\sqrt{x^2+4}}{4x} + C$

9. **Por partes** + **Sustitución algebraica**:
   $-\frac{\ln x}{\sqrt{x^2-1}} + \operatorname{arcsec}(x) + C$

10. **Sustitución** ($u = \tan x$) + **Fracciones parciales**:
    $\frac{1}{2}\ln|\tan^2 x - 1| - \ln|\tan x| + C$